

# Bayesian Compound Identification Confidence Scoring

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**Dataset:** LC-BinBase, HILIC negative-ion TTOF (Premier) **Annotated spectra evaluated:** 1,631 (manually annotated, non-yy/zz) **Unannotated spectra scored:** 5,899 (never previously queried) **Total library hits scored:** 195,367 candidate matches

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## Part I — The Problem and Approach

### 1.1 What We Are Solving

Standard LC-MS/MS library search returns a ranked list of candidate compounds with no uncertainty estimate. The top hit might be correct with 99% confidence, or it might be one of ten equally plausible structural isomers with no way to tell them apart. Analysts either accept all top hits blindly or spend large amounts of time manually reviewing individual spectra.

This project builds a **Bayesian evidence integration framework** that assigns each candidate compound a calibrated posterior probability:

$$P(\text{compound}_i \mid D) \propto p_i \cdot \text{LR}_{\text{MS2}} \cdot \text{LR}_{\text{MS1}} \cdot \text{LR}_{\text{RT}}$$

An explicit novel compound hypothesis is always included so all posteriors sum to 1:

$$\sum_i P(\text{compound}_i \mid D) + P(\text{novel} \mid D) = 1$$

When no candidate is convincingly better than the novel hypothesis, the model abstains rather than forcing a call.

Note that this is not a binary classifier — there is no labeled FP ground truth. All annotated spectra were annotated because they had good library matches. The model has no independently verified negatives. The incorrect hits from the annotated set (library candidates that are not the correct compound) serve as an empirical FP distribution for the MS2 channel, but these are a biased sample of wrong calls, not a general FP population.

### 1.2 Why Bayesian

The Bayesian framework was chosen not for accuracy but for properties that matter scientifically:

1. **Calibrated output** — a posterior of 0.9 should mean the call is correct ~90% of the time
  2. **Channel dropout** — when RT or predicted RT is unavailable, LR\_RT = 1 and drops out silently without breaking the model
  3. **Decomposable** — each channel's contribution is explicit; ablations show exactly what is and is not working
  4. **Modular prior** — the prior can be swapped or improved independently of the likelihood terms
  5. **Directly interpretable** —  $P(\text{threonine} \mid \text{this spectrum}) = 0.976$  is a statement that a chemist can act on
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## Part II — Data

### 2.1 Spectra

**ttof+neg+hilic.csv** — 7,530 LC-MS/MS features from a HILIC negative-ion TTOF run processed through LC-BinBase, downloaded directly from MassWiki. Key fields: **wiki\_id**, **rt** (observed retention time, seconds), **precursor\_mz**, **entropy** (spectral entropy  $H$  of the MS2 spectrum).

Of these, **1,883** are manually annotated (**is\_manual\_annotated** = **True**). After excluding yy/zz-prefixed entries (BinBase placeholder annotations), **1,631** form the evaluation set. The remaining 5,899 are unannotated — never previously queried against MassWiki.

### 2.2 Library Hits — Two Sources

All library hits were fetched from MassWiki, which aggregates public and in-house spectral libraries.

**Reference library** (public: NIST23, GNPS, MassBank)

- Broad chemical coverage, variable spectral quality
- RT channel: Kong 2.0 predicted RT (**predicted\_rt\_hilic**)
- Mean entropy similarity on annotated correct hits: 0.798

**Annotation library** (in-house)

- Spectra measured on the same instrument and column
- RT channel: real measured RT offset (**anno\_delta\_rt** = observed RT – library RT)
- Mean entropy similarity on annotated correct hits: 0.989
- Higher similarity and tighter RT than the reference library

When an annotation hit exists, it typically dominates. The two sources receive separate RT likelihood parameters ( $\sigma_{\text{RT\_anno}} = 14.1$  s vs  $\sigma_{\text{RT\_ref}} = 19.1$  s) reflecting this quality difference.

### 2.3 Spectrum Quality Prior

Built from 43,294 CONFIRMED TTOF HILIC negative records in **hilic\_ttof\_all.csv** (BinBase), joined to observed spectra via SPLASH identifier.

$$q(\text{wiki\_id}) = \text{peak\_gaussian\_similarity} \times \text{peak\_pure}$$

**Why quality rather than compound identity?** 90.8% of BinBase CONFIRMED records are **unknown\_xxx** entries with no compound identity. Only 101 records (100 unique compounds) carry an InChIKey. A compound-identity prior covers fewer than 0.2% of confirmed records and is effectively flat everywhere else. For further analysis, we can incorporate matrix properties, study information for better inference.

**How quality enters the model:** Rather than differentiating between candidates, the quality weight modulates how much total probability the candidate pool receives relative to the novel hypothesis:

$$\sum_i \pi_i = q(\text{wiki\_id}) \times (1 - P_{\text{novel}}) \quad \text{unnorm\_novel} = 1 - q \times (1 - P_{\text{novel}})$$

All candidates for a given spectrum share equal weight within that budget — quality affects *whether* the model commits to a call, not *which* candidate wins. This maps naturally to the problem: a spectrum with

clean peak shape and high purity is more likely to have a genuine library match; a noisy or impure peak should abstain more readily, as they may not be a real spectrum.

**Coverage:** 3,068 of 7,530 MassWiki spectra (40.7%) have a direct BinBase SPLASH match (why? i thought they would be a subset of binbase). The remaining 4,462 receive the dataset median quality (0.717). Quality range: [0.09, 0.99], mean 0.71.

## Part III — The Scoring Model

### 3.1 MS2 Likelihood Ratio

Entropy similarity is converted to an **absolute, N-independent** likelihood ratio by comparing each hit's score against empirical TP (correct hits) and FP (incorrect hits) distributions estimated from the annotated set:

$$\lambda(H) = 1 - e^{-\alpha H}$$

$$\text{LR}_{\text{MS2}, i} = \lambda(H) \cdot \frac{f_{\text{TP}}(s_i)}{f_{\text{FP}}(s_i)} + (1 - \lambda(H))$$

where  $f_{\text{TP}}$  and  $f_{\text{FP}}$  are kernel density estimates (bandwidth = 0.05) of entropy similarity for correct and incorrect library hits respectively.

- TP are defined as: library candidates where the library compound matches the human annotation for that spectrum. Specifically, correct = True, determined by InChIKey14 match between the library hit's SMILES and the observed spectrum's SMILES. When SMILES are unavailable, it falls back to name matching (lib\_name == reference\_library\_search-name).
- FP are defined as: all other library candidates returned for those same annotated spectra where correct = False. These are compounds that MassWiki returned as candidates (passing the precursor m/z filter) but are not the right answer.

When  $H \approx 0$  (single-ion spectrum),  $\lambda \approx 0$  and  $\text{LR}_{\text{MS2}} = 1$  — *MS2 is uninformative and the model falls back to MS1, RT, and the prior. When  $H$  is large,  $\text{LR}_{\text{MS2}}$  equals the empirical TP/FP density ratio at that similarity value.*

The FP distribution from the annotated set (library candidates returned for annotated spectra that are not the correct compound) serves as the null.

| Entropy similarity | LR_MS2 (at full H) |
|--------------------|--------------------|
| 0.0                | 0.0                |
| 0.5                | 0.75               |
| 0.8                | 3.84               |
| 1.0                | 82.6               |

TP distribution: 11,301 correct hits, mean similarity = 0.755 (bimodal: annotation hits peak near 0.989, reference hits peak near 0.798). FP distribution: 21,549 incorrect hits, mean similarity = 0.375.

### 3.2 MS1 Likelihood Ratio

Gaussian on  $\Delta\text{ppm}$  with  $\sigma_M = 4.21$  ppm fitted from annotated correct hits. The denominator represents a random unrelated compound (broad null,  $\sigma = 50$  ppm):

$$\text{LR}_{\text{MS1},i} = \frac{\mathcal{N}(\Delta\text{ppm}_i; 0, 4.21)}{\mathcal{N}(0; 0, 50)}$$

TTOF is less mass-accurate than Orbitrap ( $\sigma_M \approx 4.2$  ppm vs 1–2 ppm). MassWiki pre-filters candidates by precursor  $m/z$ , so the MS1 channel is most useful for discriminating among candidates at the boundary of the mass filter, where a 5–10 ppm error is a meaningful negative signal.

### 3.3 RT Likelihood Ratio — Two Channels

**Annotation hits** (real measured RT):

$$\text{LR}_{\text{RT},i} = \frac{\mathcal{N}(\text{anno\_delta\_rt}_i; 0, 14.1, \text{s})}{\mathcal{N}(0; 0, 300, \text{s})}$$

**Reference hits** (Kong 2.0 predicted RT):  $\text{LR}_{\text{RT},i} = \frac{\mathcal{N}(\text{obs\_RT}_i - \widehat{\text{RT}}_i; 0, 19.1, \text{s})}{\mathcal{N}(0; 0, 300, \text{s})}$

If no RT is available for a candidate,  $\text{LR}_{\text{RT}} = 1$  (channel drops out silently). The 300 s null represents a compound eluting at a random position on the 5-minute HILIC method.

### 3.4 Novel Compound Term and Abstention

$$P(\text{novel} \mid D) = \frac{P_{\text{novel}}}{\sum_i \text{unnorm}_i + P_{\text{novel}}}$$

If  $P(\text{novel} \mid D) \geq 0.6$ , the model abstains.  $P_{\text{novel}} = 0.224$  is estimated from the fraction of all 7,530 spectra with no library hit.

### 3.5 Fitted Parameters

All parameters estimated from the 1,631 human annotated spectra (source of ground truth):

| Parameter                  | Value    | Method                                                                                                  |
|----------------------------|----------|---------------------------------------------------------------------------------------------------------|
| $\sigma_M$                 | 4.21 ppm | Trimmed std of $\Delta\text{ppm}$ on InChIKey14-matched correct hits                                    |
| $\sigma_{\text{RT\_anno}}$ | 14.1 s   | Std of <code>anno_delta_rt</code> on annotated annotation hits, trimmed $\pm 60$ s                      |
| $\sigma_{\text{RT\_ref}}$  | 19.1 s   | Std of <code>obs_RT</code> – <code>predicted_RT</code> on annotated reference hits, trimmed $\pm 120$ s |
| $\alpha$                   | 10.0     | MLE on annotated spectra (hits optimizer upper bound)                                                   |
| $P_{\text{novel}}$         | 0.224    | Fraction of all 7,530 spectra with no library hit                                                       |

## Part IV — Performance on Annotated Spectra

### 4.1 Overall Accuracy

**Dataset:** 1,631 annotated spectra. Correctness determined by InChIKey14 match between the called library hit's SMILES and the annotated compound's SMILES from the spectra file.

| Method                              | Top-1 Accuracy   |
|-------------------------------------|------------------|
| Raw entropy similarity rank         | 99.7%            |
| Full Bayesian model (called only)   | 96.2%            |
| Full Bayesian model (all annotated) | 96.0%            |
| Abstentions                         | 9 / 1,631 (0.6%) |

The gap between the entropy baseline (99.7%) and the model (96.2%) is small and almost entirely explained by the RT channel introducing false positives — wrong compounds with coincidentally accurate predicted RTs that outweigh correct compounds lacking RT data. The baseline's 99.7% is a ceiling artifact: annotated spectra were selected precisely because they had good library matches, so entropy similarity nearly always ranks the correct compound first by construction.

## 4.2 Calibration

The model produces monotonically calibrated posteriors — higher posterior reliably means higher accuracy:

| Posterior bin | N spectra | Observed accuracy |
|---------------|-----------|-------------------|
| > 0.90        | 846       | <b>99.1%</b>      |
| 0.70–0.90     | 399       | <b>98.5%</b>      |
| 0.50–0.70     | 179       | <b>93.3%</b>      |
| 0.30–0.50     | 115       | <b>88.7%</b>      |
| < 0.30        | 83        | <b>73.5%</b>      |

At posterior > 0.90, the model is correct 99.1% of the time on 846 spectra. This is the primary practical value of the null-adjusted MS2 LR: by comparing each hit against an absolute empirical null rather than normalising within the candidate set, the posterior reflects genuine evidence strength rather than relative rank within a particular candidate pool.

## 4.3 The Duplicate Entry Problem

**87.9% of annotated spectra have more than one correct-labeled library entry** (mean 6.0 per spectrum). The same compound appears under multiple names — all sharing the same InChIKey14 — and the model scores these as independent candidates, splitting posterior mass across correct entries.

Example: Threonine has 41 correct entries across the library. The top posterior per entry is 0.063 — which looks like low confidence — but the sum across all threonine entries is 0.98.

**InChIKey14 deduplication before scoring** would collapse all entries for the same molecular connectivity into one candidate, eliminating this posterior dilution.

Ground truth labeling uses a two-level approach: (1) InChIKey14 from SMILES — the spectra file contains SMILES for 99.9% of annotated spectra, and the hits CSV contains SMILES from MassWiki for ~90% of library entries; when both are available, InChIKey14 match is used. (2) Name string fallback — for library

entries with no SMILES, **lib\_name** is matched against **reference\_library\_search-name** from the spectra file.

## 4.4 Failure Analysis

61 of 1,622 called spectra are called incorrectly (9 abstained). Of incorrectly called spectra:

- **21 (34%)**: correct compound is rank 2 — near-tie, margin < 0.1 posterior
- **40 (66%)**: correct compound is rank 3 or lower

Representative failure patterns:

**RT channel false positive (a6ICAHB/4704)**: The annotation library has a perfect match for the correct compound — Fuc( $\alpha$ 1-2)Gal( $\beta$ 1-4)GlcNAc at entropy\_sim = 1.000, delta\_ppm = 0.000. Despite this, the model calls Lewis A trisaccharide. The failure is in the RT channel: Lewis A has a Kong 2.0 predicted RT of 129.6 s against an observed RT of 131.7 s — a 2.1 s deviation, giving Lewis A LR\_RT  $\approx$  15x. The correct compound has no predicted RT and no annotation RT, so its LR\_RT = 1 (channel drops out). The RT boost for Lewis A overwhelms the MS2 advantage of the correct compound. The model is following the evidence it has — the failure is that the correct compound lacks RT data while the wrong compound coincidentally has an accurate RT prediction.

**Duplicate entries — same compound, split posterior (a6ICAHB/4072)**: BinBase annotates this spectrum as "PE 34:2e". The annotation library has an entry "PE 34:2e" with entropy\_sim = 0.9998 and near-zero mass error — effectively a perfect match. Multiple synonymous entries for the same molecular species (PE O-18:1\_16:1, PE 34:2e, 1-Stearoyl-2-linoleoyl-sn-GPC) split the posterior mass across correct entries; since all candidates receive equal per-spectrum prior weight, the top-ranked single entry reflects only a fraction of the true compound probability. This is the duplicate-entry problem (Section 4.3) — InChIKey14 deduplication before scoring would collapse these.

**Synonym conflict (a6ICAHB/900)**: 1,2-Dioleoyl-sn-glycero-3-phosphate (wrong) scores higher than its synonyms 1-Oleoyl-L- $\alpha$ -lysophosphatidic acid and LPA 18:1 (both correct). All three refer to the same compound. InChIKey14 deduplication would collapse these and resolve the conflict.

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## Part V — Expansion to Unannotated Spectra

### 5.1 Motivation

The annotated set is inherently circular: parameters are fitted on annotated spectra, evaluation is on annotated spectra, and those spectra were themselves selected because they had good library matches. The 5,899 unannotated spectra had never been queried against MassWiki.

### 5.2 Library Hit Coverage — Correcting P\_novel

| Category                       | N     | % of 7,530 total |
|--------------------------------|-------|------------------|
| Annotated spectra              | 1,631 | 21.7%            |
| Unannotated with library hits  | 4,216 | 56.0%            |
| True no-hit (no library match) | 1,683 | 22.4%            |

## 5.3 Scoring Results

167,857 candidate hits across 4,216 unannotated spectra were scored using the same parameters fitted on the annotated set.

| Outcome                                   | N     | %     |
|-------------------------------------------|-------|-------|
| Called ( $P_{\text{novel}} < 0.6$ )       | 2,090 | 49.6% |
| Abstained ( $P_{\text{novel}} \geq 0.6$ ) | 2,126 | 50.4% |

The higher abstention rate for unannotated spectra (50.4% vs 0.6% for annotated) is expected.

Confidence distribution of called spectra:

| Posterior bin | N   |
|---------------|-----|
| > 0.90        | 247 |
| 0.70–0.90     | 294 |
| 0.50–0.70     | 390 |
| 0.30–0.50     | 549 |
| < 0.30        | 610 |

## Part VI — Limitations and Known Issues

### 6.1 Annotated Set Bias

All parameters ( $\sigma_M$ ,  $\sigma_{RT}$ ,  $\alpha$ , and the TP/FP KDE distributions for MS2 LR) are fitted on annotated spectra. Annotated spectra were confirmed because they had good library matches, so:

- **$\alpha$**  always hits the upper bound (10.0) — the annotated set provides no leverage to estimate the complexity weighting because nearly all annotated spectra have  $H > 0.5$  (where  $\lambda \approx 1$  regardless of  $\alpha$ )
- **TP distribution** is biased toward high-quality matches — the distribution of correct hits is not a random sample of all correct matches in the database
- **FP distribution** is biased toward the kinds of wrong candidates that appear alongside confirmed compounds — it may not represent the FP landscape for genuinely novel spectra

### 6.2 Annotation Library RT Offset

The median `anno_delta_rt` is +9.2 s, suggesting a systematic positive shift. This is puzzling since annotation library RTs are expected to be corrected via internal standards. The Gaussian model assumes zero-mean error; the actual distribution is biased and should be investigated.

### 6.3 Duplicate Library Entries

87.9% of annotated spectra have multiple `correct`-labeled entries (Section 4.3). InChIKey14 deduplication before scoring would collapse these into single candidates, likely pushing accuracy above 98% and making posteriors directly interpretable as per-compound probabilities.

## 6.4 $\alpha$ Is Unidentifiable

$\alpha$  always hits the upper bound (10.0) in every cross-validation fold. The annotated spectra provide no leverage to estimate spectral complexity weighting because nearly all have  $H > 0.5$  (where  $\lambda \approx 1$  regardless of  $\alpha$ ).  $\alpha$  is effectively fixed, not learned.

## 6.5 RT Channel False Positives

The RT channel is the primary source of remaining errors. When a wrong compound has a coincidentally accurate predicted RT and the correct compound has no RT data, the RT channel overrules an otherwise decisive MS2 match. One mitigation could be cap LR\_RT at a maximum value (e.g., 10 $\times$ ) to prevent any single channel from dominating.

## 6.6 High-Confidence Calls on Weak Spectral Evidence

Some HIGH\_CONF calls in the unannotated set have high posterior driven by RT and MS1 alone with very low entropy similarity ( $< 0.1$ ). Without a confirmed annotation or authentic standard, these cannot be verified.